# National University
*of Computer and Emerging Sciences*
*Chiniot-Faisalabad Campus*

## NS1001-Applied Physics

# Applied Physics – Assignment 3

Chapter 21 : Coulomb's law
Practice Problems : 3, 5, 7, 10, 17, 24, 25
Chapter 22 : Electric Fields
Practice Problems : 1, 4, 5, 7, 9, 18

## Chapter 21 (solutions)

### Q3 (solution):

$q_1 = 26.0\ \mu C$
$q_2 = -47.0\ \mu C$
$Force = 5.70 N$
$r = ?$

$$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$$5.70 = (8.99\times10^9) \cdot \frac{(26\times10^{-6})(47\times10^{-6})}{(r)^2}$$

$$\sqrt{\cancel{r^2}} = \sqrt{\frac{(8.99\times10^9)\cdot(26\times10^{-6})(47\times10^{-6})}{(5.70)}}$$

$r = 1.39\ m$ (radius) Ans.

### Q5 (solution):

$q_1 = 3\times10^{-6} C$
$q_2 = -1.5\times10^{-6} C$
$r = 12\ cm \div 1000 = 0.12\ m$
$Force = ?$

$$F = K\frac{q_1 \cdot q_2}{r^2}$$

$$= (8.99\times10^9)\cdot\frac{(3\times10^{-6})(1.5\times10^{-6})}{(0.12)^2}$$

$$= 2.81\ N$$

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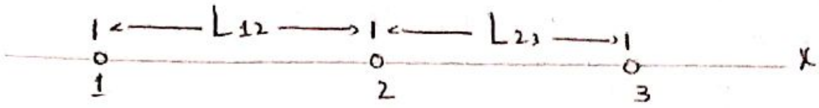
Q7 (Solution):




$$F_{net} = F_{12} + F_{23}$$

$$0 = \frac{kq_1q_3}{r^2} + \frac{kq_2q_3}{r^2}$$

$$0 = \frac{kq_1q_3}{(2L)^2} + \frac{kq_2q_3}{(L)^2}$$

$$0 = \frac{kq_1q_3 + 4kq_2q_3}{4L^2}$$

$$0(4L^2) = kq_1q_3 + 4kq_2q_3$$

$$0 = kq_1q_3 + 4kq_2q_3$$

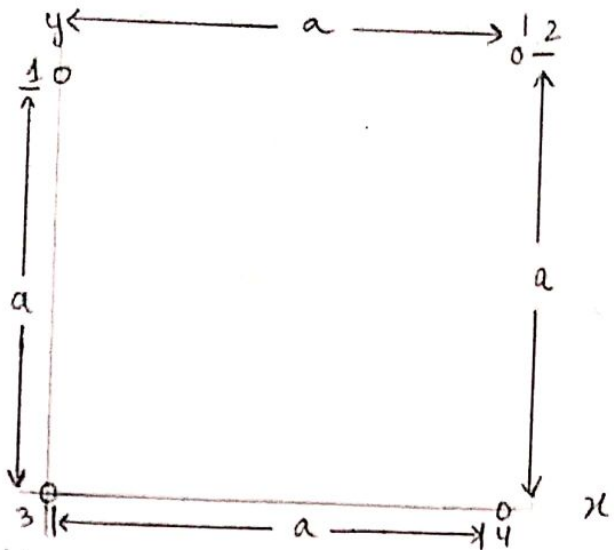
$$kq_1q_3 = -4kq_2q_3$$

$$q_1 = \frac{-4\cancel{k}q_2\cancel{q_3}}{\cancel{k}\cancel{q_3}}$$

$$q_1 = -4q_2$$

$$\frac{q_1}{q_2} = -4 \quad \text{(Ans)}$$

Q 10 (solutions):




$\therefore q_1 = q_4 = Q$

$\therefore q_2 = q_3 = q$

Part a. solution:

$$\Sigma F x = 0$$

$$\Sigma F x = \frac{k q_1 q_2}{r^2} - \frac{k Q_1 q_4 \cos 45}{r^2}$$

$$0 = \frac{k Q q}{a^2} - \frac{k Q Q \cos 45}{(a\sqrt{2})^2}$$

$$0 = \frac{2 k Q q - k Q Q \cos 45}{2a^2}$$

$$\cancel{K} Q \cancel{Q} \cos 45 = 2 \cancel{K} \cancel{Q} q$$

$$\frac{Q}{q} = \frac{2}{\cos 45}$$

$$\frac{Q}{q} = -2.83$$

Part b. solution:

$$F_{net}\, x = F_{34} + F_{32} \cos 45$$

$$0 = \frac{K q Q}{a^2} + \frac{k q q \cos 45}{2a^2}$$

$$0 = \frac{2 k q Q + k q q \cos 45}{2a^2}$$

$$0 = 2 k q Q + k q q \cos 45$$

$$-2KqQ = KQq\cos 45$$

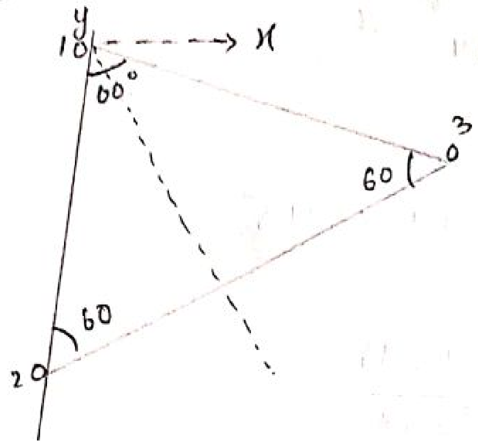
$$\frac{Q}{q} = \frac{\cos 45}{-2}$$

$$\frac{Q}{q} = \frac{-1}{2\sqrt{2}}$$

$$= 0.353$$

Ans : NO

Q 17 (solution):




$$F_1 \text{ net } x = F_{13}\sin 60 \quad (1)$$

$$= \frac{k q_1 q_3 \sin 60}{(1.5)^2}$$

$$= \frac{(8.99\times10^{9})(20\times10^{-6})(20\times10^{-6})\sin 60}{(1.5)^2}$$

$$= -0.5\text{N}$$

$$F_1 \text{ net } y = F_{12} + F_{13}\cos 60 \quad (2)$$

$$= \frac{k q_1 q_2}{r^2} + \frac{k q_1 q_3}{r^2}\cos 60$$

$$= \frac{(8.99\times10^{9})(20\times10^{-6})(20\times10^{-6})}{(1.5)^2} + \frac{(8.99\times10^{9})(20\times10^{-6})(20\times10^{-6})\cos 60}{(1.5)^2}$$

$$= 1.5982 + 0.7991$$

$$= 2.3973\text{ N}$$

Case 2 ($q_1 = q_2$)

$$F = \sqrt{(0.5)^2 + (2.3973)^2}$$

$$= 2.448\ N \text{ Ans.}$$

Q24 (solution):

$\therefore q = -1 \times 10^{-16}\ C$

$\therefore r = 1\ cm \div 100 = 0.01\ m$

Part a. solution:

$$F = \frac{k q_1 q_2}{r^2}$$

$$= \frac{(8.9 \times 10^9)(1 \times 10^{-16})(1 \times 10^{-16})}{(0.01)^2}$$

$$= 8.9 \times 10^{-19}\ N \text{ (Ans)}$$

Part b. solution:

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$$n = \frac{|q|}{e}$$

$$= \frac{-1. \times 10^{-16}}{1.6 \times 10^{-9}}$$

$$n = 625 \text{ (Ans)}$$

Q25 (solution):

$\therefore q = +1 \times 10^{-7}\ C$

$n = ?$

$$n = \frac{|q|}{e}$$

$$= \frac{1.0 \times 10^{-7}}{1.6 \times 10^{-19}}$$

$$= 6.3 \times 10^{11}$$

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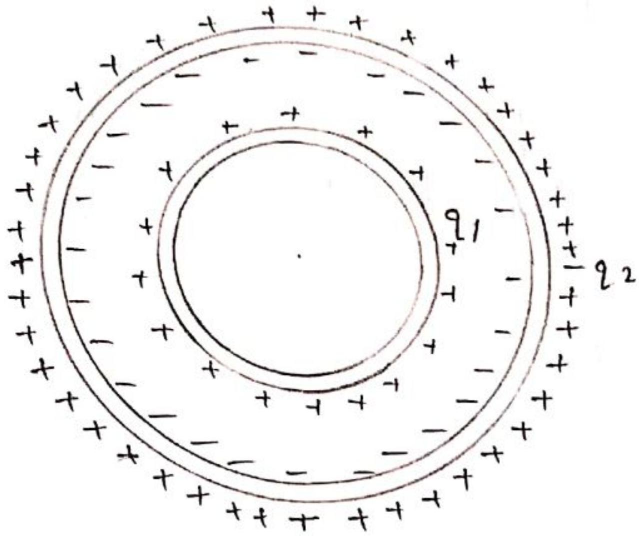
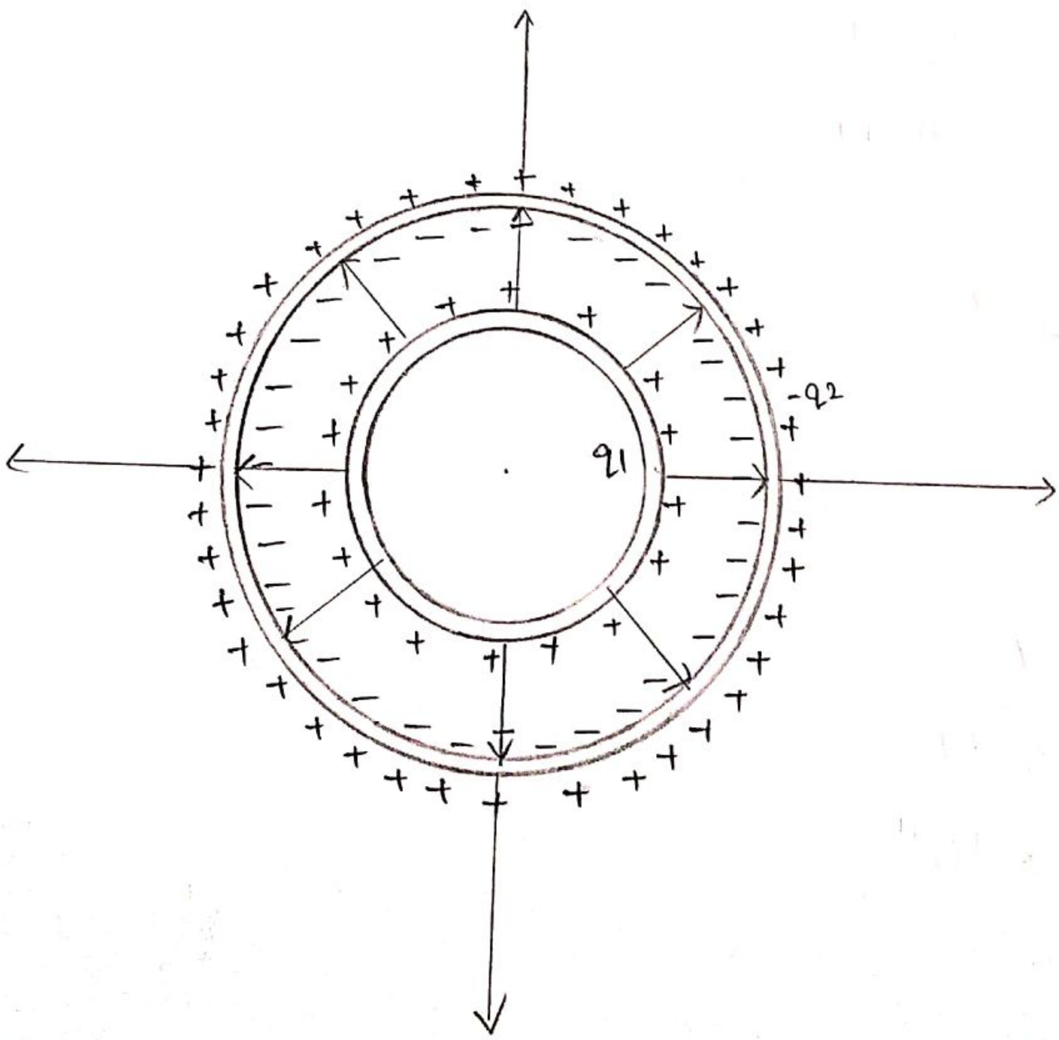
## chapter 22 (solutions)

### Q1 (solution):

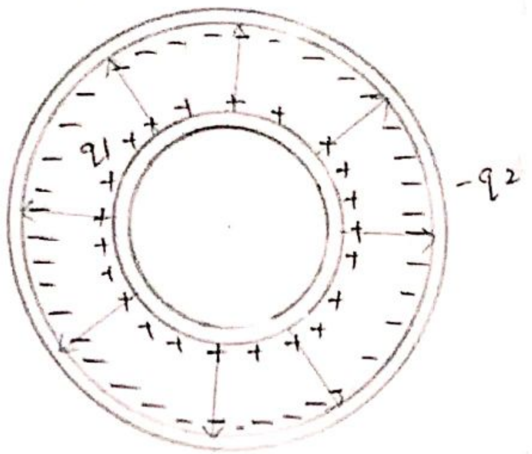
$(-q_2 + q_1)$



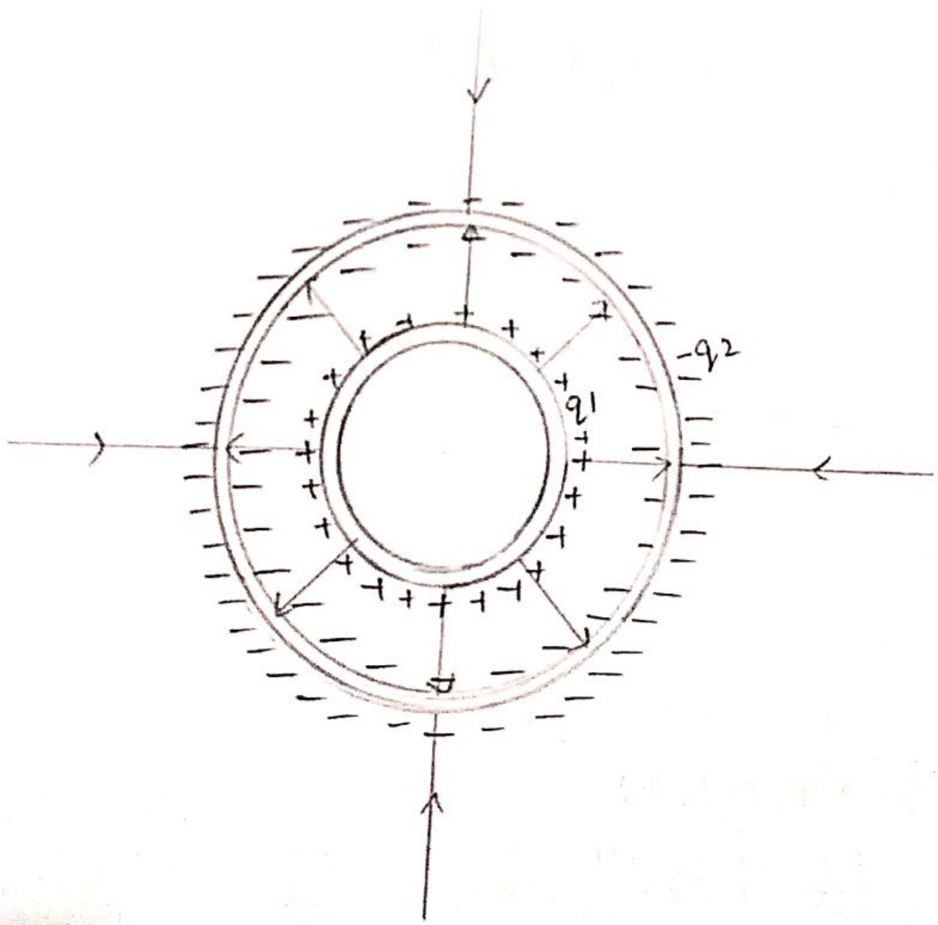

### case 1 ($q_1 > q_2$)



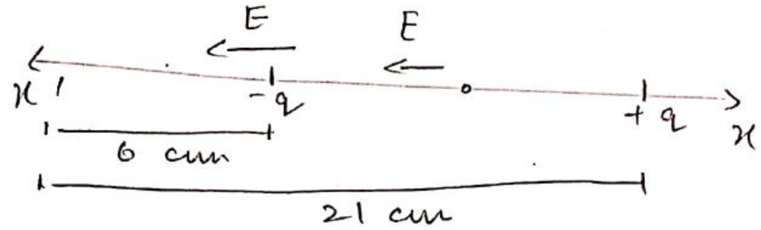
case 2 ($q1 = q2$)




case 3 ($q1 < q2$)



Q4 (solution):




$$q = 2 \times 10^{-7}\ C$$

$$x = \frac{21-6}{2}$$

$$= 7.5\ cm$$

$$E\ net = 2E$$

$$= 2 \times \frac{Kq}{x^2}$$

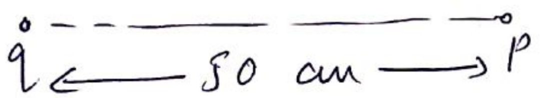
$$= \frac{2 \times (8.99 \times 10^9)(2 \times 10^{-7})}{(7.5 \times 10^{-2})^2}$$

$$E\ net = 6.4 \times 10^5\ N/C$$

Q5 (solution):

$$E = \frac{Kq}{r^2}$$

$$E = 2\ N/C$$




$$E = \frac{Kq}{r^2}$$

$$q = \frac{E r^2}{K}$$

$$= \frac{2 \times (50 \times 10^{-2})^2}{(8.99 \times 10^9)}$$

$$q = 5.56 \times 10^{-11}\ C$$

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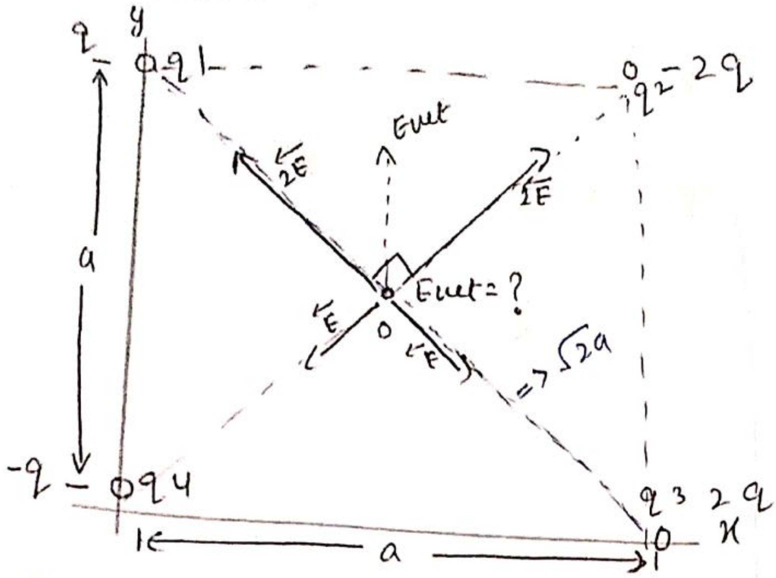
Q7 (solution):




$q_1 = q = +10.0\,nC$

$q_2 = -2q = -20.0\,nC$

$q_3 = 2q = +20.0\,nC$

$q_4 = -q = -10.0\,nC$

$a = 5.00\,cm = 5 \times 10^{-2}\,m$

$\therefore$ diagonal $= \frac{\sqrt{2}\,a}{2}$ = radius $= \frac{a}{\sqrt{2}}$

$E_{net} = \sqrt{E^2 + E^2}$

$= \sqrt{2E^2}$

$= \sqrt{2}\,E$

$= \sqrt{2} \times 7.2 \times 10^4$

$= 1.02 \times 10^5\,N/C$

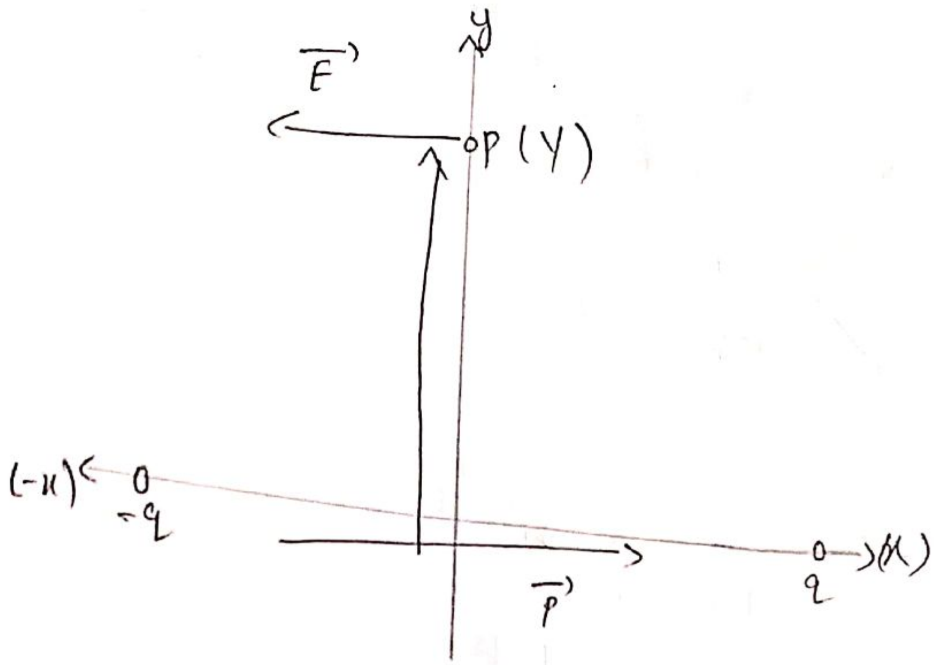
$\Longrightarrow$

$E = \frac{kq}{r^2}$

$= \frac{(8.99 \times 10^9)(10 \times 10^{-}}{\left(\frac{5 \times 10^{-2}}{\sqrt{2}}\right)^2}$

$= 7.2 \times 10^4\,N/C$

Final Answer: $E_{net} = 1.02 \times 10^5\,N/C$




$$q = 3.2 \times 10^{-19}\ C$$

$$x = 3\ m$$

$$y = 4\ m$$

$$E_{eq} = \frac{kp}{(r^2 + d^2)^{3/2}}$$

$$= \frac{kp}{(r^2 + x^2)^{3/2}}$$

$$= \frac{(8.99 \times 10^9)\, q\, 2x}{(y^2 + x^2)^{3/2}}$$

$$= \frac{(8.99 \times 10^9)(3.2 \times 10^{-19})(2 \times 3)}{(4^2 + 3^2)^{3/2}}$$

$$= 1.38 \times 10^{-10}\ N/C$$

$$\vec{E}_p = -(1.38 \times 10^{-10}\ N/C)\uparrow$$ Ans.

Q 18 (solution):

$$E = \frac{1}{2\pi\varepsilon_0} \cdot \frac{qd}{z^3} + E_{next} + \cdots$$

Find $E_{next}$ ?

$$E = \frac{q}{2\pi\varepsilon_0 z^3} \cdot \frac{d}{\left[1-\left(\frac{d}{2z}\right)^2\right]^2} + \cdots$$

Binomial expansion.

$$E = \frac{qd}{2\pi\varepsilon_0 z^3}\left[1-\left(\frac{d}{2z}\right)^2\right]^2$$

$$= \frac{qd}{2\pi\varepsilon_0 z^3}\left[1+\left\{-\left(\frac{d}{2z}\right)^2\right\}\right]^{-2}$$

$$= \frac{qd}{2\pi\varepsilon_0 z^3}\left[1+(-2)\left\{-\left(\frac{d}{2z}\right)^2\right\}\right]^{-2}$$

$$= \frac{qd}{2\pi\varepsilon_0 z^3}\left[1+\frac{{}^{1}\cancel{2}d^2}{{}_{2}\cancel{4}z^2}+\cdots\right]$$

$$= \frac{qd}{2\pi\varepsilon_0 z^3} + \frac{qd^3}{4\pi\varepsilon_0 z^5}$$

$$= \frac{1}{2\pi\varepsilon_0 z^3} \cdot qd + \frac{1}{4\pi\varepsilon_0} \frac{qd^3}{z^5} + \cdots$$

$$\therefore E_{next} = \frac{1}{4\pi\varepsilon_0} \frac{qd^3}{z^5} \text{ (Ans)}$$

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